

- Increased capacity in the EU R&I system to conduct open science and to set it as a *modus operandi* of modern science.

Scope: Monographs, books and other types of long-text formats are very important outputs in some disciplines, such as in the Social Sciences and the Humanities. Whilst many disciplines rely nearly exclusively on the production of articles, professional advancement requirements, as well as the nature of the research and the writing process in some disciplines, result in the unwavering significance of long-text formats and monographs, both in the digital and in the print formats.

Open access for monographs and other long-text formats has been a slow process in multiple aspects. This is likely because of the higher costs and business models of book publishing, as well as a hesitancy on the side of researcher to trust important parts of their work in terms of volume to the new open access publishing environment. Additionally, funder policies and strategies focus a lot less on monographs than on journal articles, and they are often recommendations and not requirements.

This action aims to support the development of aligned institutional and funder policies and strategies regarding open access to books and monographs in Europe. In this context, the action should support: 1. activities that foster greater understanding of bottlenecks in the development of policies and strategies for open access to books and monographs by institutions and funders; 2. the coordination of policies amongst funders and research institutions, and the sharing of good practices in policies and strategies for open access to books and monographs; 3. actionable and evidence-based recommendations and resources for comprehensive institutional and funder policies for open access to books and monographs. The action may also envisage international cooperation with entities outside the EU Member States and associated countries.

The action should be no longer than 24 months.

HORIZON-WIDERA-2022-ERA-01-44: Developing and piloting training on the practice of open and responsible research and innovation

Specific conditions	
<i>Expected EU contribution per project</i>	The Commission estimates that an EU contribution of around EUR 3.50 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.

⁸⁴ Open science practices may include: providing open access to research outputs (such as publications, data, software, models, algorithms, and workflows); early and open sharing of research (for example through preregistration, registered reports, pre-prints, crowd-sourcing of solutions to a specific problem); participation in open peer-review; measures to ensure reproducibility of results; and involving citizens, civil society and end-users in the co-creation of R&I agendas and content, including citizen science.

<i>Indicative budget</i>	The total indicative budget for the topic is EUR 3.50 million.
<i>Type of Action</i>	Coordination and Support Actions
<i>Legal and financial set-up of the Grant Agreements</i>	The rules are described in General Annex G. The following exceptions apply: Beneficiaries will be subject to the following additional dissemination obligations: Beneficiaries must make proactive efforts to disseminate policy recommendations aimed at relevant research institutions and levels of governance. Beneficiaries must acknowledge and incorporate this obligation in the proposal, outlining the efforts they will make towards meeting them, and in Annex I to the Grant Agreement.

Expected Outcome: Projects are expected to contribute to the following expected outcomes:

- Consolidated evidence base on open and responsible research and innovation⁸⁵ training for researchers at all stages of their careers across the European Research Area.
- Development and piloting of high-quality training to a significant number of researchers on diverse aspects of open and responsible research and innovation.
- Establishment of a platform for further development of training, mutual learning, and dissemination of recommendations to authorities and institutions responsible for researcher training.

These targeted outcomes in turn contribute to medium and long-term impacts:

- Improved skills leading to greater employability and career prospects for researchers.
- Improvements in the excellence of the science conducted, improved capacity within the European Research Area to tackle societal challenges, greater interaction between science and society, improved overall effectiveness of EU research and innovation funding.
- Improved engagement and cooperation of researchers, with communicators, journalists, industry, policy maker and civil society.

Scope: In order to ensure the excellence of EU research and innovation, the research workforce needs to strengthen skills related to integrity, avoidance of various forms of bias, including gender bias, non-discrimination and inclusion, FAIR data management, data robustness and reproducibility, dissemination and exploitation of results, and interdisciplinary research. It requires skills to communicate science and research results, in the media (including social media) and to policy makers, and skills to participate in public debates. It also requires skills to engage the public to develop research and innovation agenda, the ability

⁸⁵ Responsible research and innovation involves multi-actor and public engagement in research and innovation, easier access to scientific results, the take up of gender and ethics in the research and innovation content and process, and formal and informal science education.

to work in multi-actor collaborations covering all parts of the quadruple helix⁸⁶ where interests and motivation for engaging in research may be diverse, and actively involve citizens in research activities such as citizen science to expand the scope and the potentials of science itself. These skills are needed at all stages of researchers' careers, from undergraduate onwards (including as part of doctoral training).

This action consists of three parts, all of which must be addressed:

The first part consists of consolidating existing evidence⁸⁷ on researcher training on open and responsible research and innovation, at all stages of their careers, from undergraduate onwards (including as part of doctoral training), and identifying gaps in existing practices in the European Research Area. This will include mutual learning activities within and outside the European Research Area to ensure established and emergent good practices are considered. Existing initiatives, such as the European Charter and Code⁸⁸ and the EU Principles for Innovative Doctoral Training⁸⁹, should be taken into account.

The second part consists of developing training modules, courses or other kinds of interventions on open and responsible research and innovation that will be piloted to a significant number of students and researchers in different kinds of research organisations and higher education organisations across Europe, and at all stages of their studies and careers. The learning outcomes will be evaluated, with a view to refining the training ready for more widespread deployment (e.g. at national, institutional levels, or in MSCA actions). Attention should be paid to the gender dimension of the training/learning outcomes, and to ensure that the training is adapted and piloted to researchers of all backgrounds (age, ethnicity, geography, gender, etc.).

The third part consists of establishing a platform, sustainable beyond the lifetime of project funding, to continue knowledge exchange, develop guidance, and further develop training on open and responsible research and innovation. Part of the work must include the wide dissemination of policy recommendations aimed at relevant research institutions and levels of governance.

Working closely with relevant on-going actions, quadruple helix stakeholders, and science outreach stakeholders, is advised. The action is also expected to build on and valorise results from projects already funded under the "Science with and for Society" part of Horizon 2020.

HIGHER EDUCATION INSTITUTIONS AND RESEARCHERS

Proposals are invited against the following topic(s):

⁸⁶ A model of cooperation between industry, academia, civil society and public authorities, with a strong emphasis on citizens and their needs.

⁸⁷ In addition to relevant national-level initiatives, or even initiatives outside the EU, this includes Horizon 2020 projects supported through topics SEAC-2-2014 and topic SwafS-08-2019-2020 under Science with and for Society; the Horizon 2020 Innovative Training Networks supported through the MSCA-ITN topics as well as the COFUND projects for doctoral programmes under the Marie Skłodowska-Curie Actions, and the Erasmus+ European University Alliances and Horizon 2020 top-up pilot.

⁸⁸ <https://euraxess.ec.europa.eu/jobs/charter>

⁸⁹ <https://www.euraxess.be/belgium/jobs-funding/doctoral-training-principles>